

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LAILA | Context: Arab-Country G10–12 Arabic Teachers: AI Essay Grading Assistance

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology consists of seven trait scores plus a holistic sum [Q34, Q35], with six traits on a 0–5 scale and REL on a 0–2 scale [Q36], grounded in the QUTC CAST rubric [Q33]. Three concerns drive the low score, two of which are flagged HIGH priority in elicitation. (1) The trait taxonomy is Qatari-curriculum-derived; no mapping is documented to Egyptian holistic grading [WEB-5], Tawjihi [WEB-10], Qiyas [WEB-7], CRDP, or Moroccan frameworks. Egypt's holistic-only practice means the analytic seven-trait label space may not be the construct teachers actually evaluate (elicitation Q2, Q4). (2) The output is exclusively discrete point-score labels [Q181, Q182]; the deployment requires confidence and rationale outputs [elicitation Q3], which are entirely absent from the output ontology. (3) The narrow REL 0–2 scale is acknowledged by the authors as limiting QWK granularity [Q91]. Strengths: the trait set is internally coherent, the REL=0 cascade rule [Q37] is explicit, and the rubric is documented in both Arabic [Q114] and English [Q116, Q118]. But for teacher-facing deployment across multi-country contexts, the output ontology is structurally misaligned on two of the highest-priority elicitation dimensions.

ID	Evidence
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q35	"Additionally, a Holistic score (HOL) was computed as the sum of the 7 trait scores." (p.5)
Q36	"Six traits (all but REL) were rated on a 6-point scale (0 = lowest, 5 = highest), while REL was rated on a 3-point scale (0 = not relevant, 1 = partially relevant, and 2 = fully relevant)." (p.5)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q2	"The dataset comprises 7,859 essays across 8 distinct prompts making it the most extensive publicly available Arabic resource of its kind." (p.1)
Q4	"Table 1: Brief description of the traits in LAILA." (p.1)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q182	"The holistic score was computed as the sum of the individual trait scores." (p.18)
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)
Q37	"Furthermore, if an essay received a REL score of 0, all other trait scores were set to 0, as irrelevant responses are not subject to further evaluation." (p.5)
Q114	"For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC)." (p.12)
Q116	"An English-translated version of the CAST rubric is presented in Table 6." (p.12)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)

WEB-10	Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition (https://en.wikipedia.org/wiki/Tawjihi)
WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)

Strengths

- Seven-trait analytic schema aligns with rubric-based grading practice in Gulf countries (Qatar, UAE, Saudi Arabia) [Q34, WEB-7]
- Explicit rubric documentation (Arabic original + English translation) supports interpretability [Q114, Q116, Q118]
- Clear scoring scale and decision rules including REL=0 cascade [Q36, Q37]
- Score distributions reported per trait per prompt give deployment users a sense of label calibration [Q40, Q119]

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Q36	"Six traits (all but REL) were rated on a 6-point scale (0 = lowest, 5 = highest), while REL was rated on a 3-point scale (0 = not relevant, 1 = partially relevant, and 2 = fully relevant)." (p.5)
Q37	"Furthermore, if an essay received a REL score of 0, all other trait scores were set to 0, as irrelevant responses are not subject to further evaluation." (p.5)
Q40	"Most traits follow a near-normal pattern on the 6-point scale, indicating overall positive essay quality. The REL scores show that 86% of the essays agree strongly with the prompt. In contrast, GRA and VOC show the highest percentage of score 1 at 16%, followed by MEC at 13%, highlighting that these traits pose the greatest challenges." (p.6)
Q119	"Figure 4 presents the score distributions in LAILA across 8 distinct prompts (P1–P8) and 7 writing traits, including Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA), alongside a Holistic (HOL) score representing the sum of all individual traits." (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Categories are regionally relevant for analytic-rubric users (Gulf, UAE) but not for holistic-grading users (Egypt) [WEB-5]; cross-national rubric mapping is not documented [WEB-7, WEB-10].

OO-2: Missing categories for the deployment: (1) no holistic-only output mode for Egyptian-style grading; (2) no confidence/uncertainty category despite elicitation Q3 requiring it; (3) no rationale/justification category despite elicitation Q3 requiring it; (4) no narrative/descriptive trait specifications since those genres are excluded [Q89].

OO-3: The CAST rubric is QUTC-derived and may encode Gulf-curriculum stylistic and developmental norms in Style, Development, and Organization scoring; the elicitation flags this as warranting verification (Q1, OC priority) rather than dismissal. No web evidence documents how CAST descriptors compare with Maghrebi or Levantine rhetorical traditions.

OO-4: Stakeholder-driven taxonomy redesign is recommended but not undertaken in LAILA; the dataset's own ethics statement requires fairness auditing and stakeholder consultation before real-world application [Q106].

OO-5: Documented taxonomy issues harming structural and content validity: (1) discrete-label-only output form omitting confidence and rationale [Q181, Q182]; (2) narrow REL scale [Q91]; (3) no cross-national rubric mapping [WEB-5, WEB-7, WEB-10]; (4) no holistic-only mode [WEB-5].

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WEB-5	Egypt's holistic grading culture suggests Egyptian teachers may apply different evaluative standards than Qatar-based annotators (https://grokipedia.com/page/academic_grading_in_egypt)
WEB-7	Saudi Qiyas focuses on higher-education entry testing rather than secondary multi-trait classroom rubrics (https://etec.gov.sa/en/centers/qiyas)
WEB-10	Jordan's Tawjihi Arabic exam emphasizes literary comprehension and grammar over trait-scored free composition (https://en.wikipedia.org/wiki/Tawjihi)
Q89	"Second, LAILA includes only explanatory and persuasive writing prompts, limiting genre diversity and potentially affecting model robustness across other styles, such as narrative or descriptive writing." (p.10)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q182	"The holistic score was computed as the sum of the individual trait scores." (p.18)
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q1	"LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar." (p.1)
Q106	"We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application." (p.10)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)

Information Gaps

- Whether CAST trait descriptors carry Gulf-specific stylistic norms detectable in score distributions
- Whether any Arab country's national rubric maps onto a subset of the seven traits

Requires Expert Verification

- Whether Egyptian, Lebanese, or Moroccan teachers would re-weight or re-define Style and Development categories given their professional norms

Remediation

Gap 1: No category for confidence/uncertainty or rationale outputs, despite both being HIGH-priority deployment requirements [elicitation Q3]

Recommendation: Extend the output ontology with two new categories: (1) per-trait confidence/uncertainty signal, and (2) rationale text. Adapt an English rationale-generation framework (RaDME [WEB-13] or RMTS [WEB-12]) to Arabic with

teacher-facing evaluation.

ID	Evidence
Q3	“Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings.” (p.1)
WEB-13	RaDME generates trait scores with co-produced rationales using knowledge distillation — English only, no Arabic adaptation documented (https://arxiv.org/html/2502.20748v1)
WEB-12	BERT-based Arabic AES system documented but without rationale generation (https://www.researchgate.net/publication/382302599_Automated_essay_scoring_in_Arabic_a_dataset_and_analysis_of_a_BERT-based_system)